National University of Technology

Department of Artificial Intelligence

Swarm Intelligence

Final Phase

Comparative Analysis of Phases 1–4

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1. Introduction

Across Phases 1–4 of this course project, the group studied a wide collection of research papers centred on **Opposition-Based Learning (OBL)** and its integration with **Particle Swarm Optimization (PSO)** and related metaheuristics. The focal paper of the project is *Opposition Based Initialization in Particle Swarm Optimization (O-PSO)* by Jabeen, Jalil, and Baig (GECCO 2009). The other papers were chosen because they are direct intellectual neighbours of O-PSO — predecessors that introduced OBL, contemporaries that proposed alternative OBL placements in PSO, successors that extended OBL into other PSO variants and metaheuristics, and a survey that puts the whole research line in context.

This final report has two purposes:

1. List the **pros and cons** of each paper studied.
2. Identify **which paper is the strongest contribution** of the set, with justification.

The papers are grouped into seven thematic categories so the comparison stays readable.

2. Methodology of Comparison

The papers are evaluated on four criteria:

- **Methodological rigour** — statistical tests, ablations, fair function-evaluation budgets, control experiments.
- **Novelty of contribution** — originality of the proposed mechanism beyond prior OBL/PSO work.
- **Reproducibility** — clarity of pseudocode, parameter values, and dataset details.
- **Practical impact** — breadth of follow-up work and applicability beyond synthetic benchmarks.

These four criteria are revisited explicitly in Section 6 when selecting the strongest paper overall.

3. The Focal Paper: O-PSO (Jabeen, Jalil, Baig, 2009)

O-PSO seeds the swarm with both a random population and its opposite, retains the best N of the combined $2N$ candidates, and then runs standard PSO with Clerc constriction. It is positioned as a lightweight, plug-in improvement over random initialization. It is simple, influential, and easy to extend — but its experimental footprint is modest (only four benchmarks, swarm size 20, no statistical significance tests). The papers in Phases 2–4 either replace random initialization with a smarter scheme, move OBL deeper into the PSO evolution loop, or transplant OBL to other metaheuristics. The comparison below traces how the OBL idea matured in the years after O-PSO.

4. Pros and Cons of Each Paper Studied

4.1 Group A — Foundational Work

A1. Tizhoosh (2005) — A New Scheme for Machine Intelligence

The seminal paper that introduced Opposition-Based Learning as a general-purpose machine-intelligence scheme.

Pros:

- Conceptually elegant and trivial to add to any existing optimization algorithm.
- Highly general — applicable to GAs, neural networks, reinforcement learning, PSO, and DE.
- Seeds an entire research line that grew through 2005 to the present.

Cons:

- Each application doubles fitness evaluations at the points where OBL is invoked.
- Uses a purely linear opposite formula that does not always match the geometry of the search space.
- Improvement is not guaranteed in highly non-linear or constrained problems — often needs hybridisation.

4.2 Group B — OBL Applied at Initialization

B1. Jabeen, Jalil, Baig (2009) — O-PSO (Focal Paper)
Random and opposite candidates at initialization, then standard PSO.

Pros:

- Simple plug-in modification that can be added to any PSO implementation.
- Includes a clear control experiment (2-PSO) that isolates opposition from simply doubling the random pool.
- Shows consistent improvement on three of four benchmarks.

Cons:

- Only four benchmark functions; no rotated or shifted CEC tests.
- No statistical significance tests despite 200 runs.
- Small swarm (20) and limited dimensionality (10–30) in the main setup.
- No function-evaluation budget analysis — hides the cost of doubled initial evaluations.
- No theoretical justification for why opposition should work.

B2. Tian et al. — Chaotic PSO
Replaces random initialization with chaotic-map-driven sampling.

Pros:

- Improves diversity at the start and reduces premature convergence.
- Simple modification with noticeable performance gains.

Cons:

- Performance depends heavily on which chaotic map is used.
- Parameter-sensitive and slightly more complex than standard PSO.
- Limited testing on high-dimensional problems.

B3. Tian et al. — LatinPSO (Latin Hypercube Sampling)
Uses LHS for initialization while simultaneously inferring ODE structure and parameters.

Pros:

- Stratified sampling gives uniform coverage of the search space.
- Targets a real-world application (biological ODE inference) rather than only benchmarks.

- Faster convergence and better accuracy than standard PSO.

Cons:

- More complex to implement than standard PSO.
- Higher computational cost.
- Limited comparison set; scalability to very high dimensions unclear.

B4. Kannan & Diwekar — Enhanced PSO with Quasi-Random Numbers
Uses Sobol/Halton sequences for an evenly distributed initial population.

Pros:

- Better space coverage than pseudo-random initialization.
- Faster convergence and more accurate results.
- Lower chance of getting trapped in local optima.

Cons:

- Performance depends on the chosen low-discrepancy sequence.
- Slight implementation overhead.
- Evaluation limited to benchmark problems.

B5. Gao, Liu, Huang — CSPSO (Chaotic OBL Init + Stochastic Search)
Combines chaotic OBL initialization with a stochastic search step in the main loop.

Pros:

- More diverse and evenly distributed starting population.
- Stochastic search helps escape local optima.
- Tested on 30 to 300 dimensions, showing some scalability.

Cons:

- Two compounding mechanisms make the algorithm harder to understand.
- Extra computational cost in both initialization and search phases.
- Results sensitive to chaotic-map choice and stochastic factors.

B6. Bangyal et al. — WE-PSO (WELL-based Initialization)
Uses the WELL random-number generator for a better-distributed initial swarm.

Pros:

- More uniform initialization than standard PRNGs.
- Gains grow with problem dimensionality.
- Effective on benchmarks and ANN training tasks.

Cons:

- Comparison set is mostly PSO variants and excludes modern non-PSO optimizers.
- Superiority not consistent across all benchmark functions.
- The isolated contribution of just WELL is not cleanly ablated.

4.3 Group C — OBL Applied Throughout the Search

C1. Wang, Liu, Li, Zeng (2007) — OPSO with Cauchy Mutation
OBL applied probabilistically every generation, plus Cauchy mutation on the global best.

Pros:

- Two complementary mechanisms (diversity + escape from local optima).
- Dynamic search bounds keep opposition relevant as the swarm contracts.
- Partial ablation via OPSO-noCauchy and PSO-Cauchy baselines.

Cons:

- Only 50 runs and no Wilcoxon or Friedman significance tests.
- Jumping rate J_r fixed without sensitivity analysis.
- Weak baseline (only PSO-IW); no scaling beyond 30 dimensions.

C2. Shahzad et al. (2009) — OVCPSO
OBL throughout the search, plus velocity clamping and decreasing inertia weight.

Pros:

- OBL applied during the entire search rather than just initialization.
- Uses Number of Function Calls (NFC) as a fair performance metric.
- Combines three complementary mechanisms targeting different PSO failure modes.

Cons:

- J_r not adaptive or analysed.
- Benchmark suite omits rotated and shifted CEC functions.
- No ablation isolating the contribution of OBL, clamping, or inertia schedule.
- Velocity not reset after an OBL jump (theoretical inconsistency).

C3. Wang, Wu, Rahnamayan, Liu, Ventresca — GOPSO (Generalized OBL + Cauchy)
Generalises OBL to a larger family of opposite points and adds Cauchy mutation.

Pros:

- Improved exploration–exploitation balance.
- Faster convergence and better optima on large-scale tests.
- Cauchy mutation explicitly helps escape multimodal traps.

Cons:

- Cost roughly doubles since both original and transformed solutions are evaluated.
- Gains not guaranteed in dynamic or very complex environments.
- Real-world deployment not demonstrated.

C4. Liu, Xu, Wang, Zeng (2015) — BPSO-OBL
Applies OBL to personal-best positions and introduces a "rebel learning" repulsive term from the worst pbest.

Pros:

- Most rigorous evaluation in this research line — Wilcoxon tests, seven competitors including strong baselines (CLPSO, GOPSO, FIPS), scaled to 50 dimensions.

- Genuinely novel mechanism: targets the root cause of BPSO's premature convergence rather than patching it.
- Principled design decision to omit initialization OBL, justified by evaluation budget reasoning.

Cons:

- Rebel-learning weight α drawn randomly without sensitivity analysis.
- Dynamic bounds for pbest opposition can be skewed by outlier particles.
- "Parameter-free" claim is somewhat overstated.

C5. Wang (2012) — OBPSO (Barebones PSO + OBL for Constrained Problems)
Adds OBL, a boundary-search strategy, and an adaptive penalty method to Barebones PSO.

Pros:

- Targets constrained nonlinear problems — closer to engineering reality.
- Boundary search is theoretically motivated since constrained optima often lie on boundaries.
- Adaptive penalty removes manual tuning of penalty coefficients.

Cons:

- No comparison against dedicated constrained optimizers (ε-DE, CMA-ES).
- Dated benchmark set (g01–g13); newer CEC constrained suites available.
- No statistical significance testing; boundary-search procedure underspecified.

4.4 Group D — OBL in PSO Variants with Specific Innovations

D1. Wu, Ni, Zhang, Gu (2008) — OCLPSO
Generates opposite exemplars during the per-dimension learning step of Comprehensive Learning PSO.

Pros:

- Novel integration point — OBL inside exemplar selection rather than at init or via generation jumping.
- Builds on CLPSO, a strong parent algorithm.
- Clean controlled comparison vs. pure CLPSO.

Cons:

- Only 30 runs, no statistical tests.
- Benchmark suite lacks shifted/rotated CEC 2005 functions.
- OBL becomes counterproductive late in search but no adaptive disabling mechanism.
- Implementation details (learning probability schedule, refreshing gap) underspecified.

D2. Too, Sadiq, Mirjalili — COPSO (Conditional OPSO for Feature Selection)
Combines a conditional strategy with OBL for feature-selection problems.

Pros:

- Explicit balance of exploration and exploitation.
- Higher classification accuracy with fewer selected features.

- Tested across multiple benchmark datasets.

Cons:

- Parameter settings fixed across all problem types.
- Small swarm size limits very high-dimensional feature spaces.
- Mostly benchmark datasets; real-world deployment limited.

D3. García-Morales et al. — IOBL-PSO

Refines the OBL formulation inside standard PSO for continuous problems.

Pros:

- Drop-in integration into the traditional PSO framework.
- Uses Wilcoxon tests, strengthening statistical claims.
- Consistent improvement on benchmarks.

Cons:

- Extra OBL operations raise runtime.
- Only PSO-family competitors — no modern non-PSO baselines.
- Theoretical explanation of why "improved" OBL is improved is thin.

D4. Bashir et al. (2017) — PSO with OBL and Near-Neighbour Interactions

Adds peer-to-peer information sharing on top of OBL.

Pros:

- Near-neighbour interaction improves information flow within the swarm.
- Better exploration–exploitation balance and reduced premature convergence.
- Easy integration into existing PSO frameworks.

Cons:

- Performance highly dependent on the neighbourhood-definition strategy.
- Only benchmark function testing.
- Limited theoretical analysis of convergence behaviour.

D5. Imran et al. — Opposition-Based PSO with Mutation Operators

Combines OBL with mutation operators to maintain diversity.

Pros:

- Mutation provides an extra escape route from local optima.
- OBL provides broad search-space coverage.
- Easy to integrate into existing PSO implementations.

Cons:

- Sensitive to the choice of mutation and opposition parameters.
- Limited experimental coverage.
- No formal convergence proof.

D6. Hassan et al. — Improved OBL-PSO for Global Optimization

Combines several OBL strategies and improved random-initialization methods inside PSO.

Pros:

- Clear accuracy and efficiency gains on benchmarks.
- Improves population diversity and avoids local optima.
- Statistically validated results.
- Demonstrated on benchmark functions and ANN training tasks.

Cons:

- Stacking several techniques makes the method harder to implement and explain.
- Higher computational complexity.
- Heavily parameter-tuned; real-world performance not fully verified.

4.5 Group E — OBL in Other Metaheuristics (beyond PSO)

E1. Rahnamayan, Tizhoosh, Salama — ODE (Opposition-Based Differential Evolution)
Adds OBL initialization and generation jumping to standard DE.

Pros:

- Clear, well-engineered integration of OBL into DE.
- Faster convergence and higher accuracy than standard DE on benchmarks.
- Maintains population diversity through generation jumping.

Cons:

- Evaluation focused on benchmarks; little real-world application.
- Cost of evaluating both candidate and opposite solutions not deeply analysed.
- Lacks practical guidance for tuning jumping rate and population size.

E2. Ahandani & Alavi-Rad — OBL in Shuffled Differential Evolution
Integrates OBL into SDE to improve diversity in high-dimensional problems.

Pros:

- SDE shuffling and OBL are genuinely complementary.
- Four design variants are compared, clarifying which choices help most.
- Tested on all 25 CEC 2005 functions with non-parametric statistical tests.
- Solves previously unsolved benchmark functions — a notable contribution.

Cons:

- Evaluation limited entirely to synthetic benchmarks.
- Effectively doubles function evaluations in OBL-intensive stages.
- No clear guidance for jumping rate, population size, or memeplex count.

E3. Cao, Li, Wang — OBL in Animal Migration Optimization
Applies OBL to a non-PSO metaheuristic (AMO) at both initialization and evolution.

Pros:

- Demonstrates that OBL generalises beyond PSO and DE.
- Outperforms AMO and several standard optimizers on 23 benchmarks.

Cons:

- AMO itself was an immature algorithm at the time, raising trust concerns.
- Results across the 23 functions are not uniformly strong.
- No statistical significance tests.

4.6 Group F — Domain-Specific Applications

F1. Kang, Xiong, Zhou, Meng — Opposition-Based Hybrid PSO for Noisy Environments

Uses probabilistic OBL to make PSO robust to additive noise.

Pros:

- Explicitly targets noisy optimization problems.
- Selects best particles across normal and opposite swarms.
- Benchmarked against several DE variants in addition to PSO.

Cons:

- All noise is artificially injected Gaussian noise — no real-world noisy problem.
- Jumping-rate parameter not fully analysed.
- OBL across the whole search is costly for expensive objective functions.

F2. Cao, Ding, Wang, Chen — Opposition-Based Improved PSO for Reactive Power Dispatch

Combines OBL with inertia weight, neighbourhood exchange, and crossover-mutation for an actual power-system control problem.

Pros:

- Solves a genuine engineering problem with a realistic three-part cost model (loss + voltage + transformer).
- Tested on both small and large power-system networks.
- Outperforms PSO, GSA, GA, and CLPSO on the studied systems.

Cons:

- Only the best run out of 50 is reported — no average or consistency measure.
- Many parameters and no tuning guidance.
- Comparison with DE and OGSA is unfair due to different variable encodings.
- The opposition concept itself is reused, not original.

4.7 Group G — Survey / Review

G1. Xu, Wang, Wang, Hei, Zhao (2014) — A Review of OBL from 2005 to 2012

Surveys 138 OBL papers across multiple metaheuristic families.

Pros:

- Broad, coherently organised taxonomy of OBL theory, variants, and applications.
- Clarifies relationships among standard OBL, quasi-opposite, quasi-reflected, and GOBL.
- Explicitly identifies open problems (adaptive J_r, non-uniform distributions, constrained settings).

- Peer-reviewed in EAAI (Elsevier).

Cons:

- Coverage stops at 2012, missing important post-2012 work.
- Uneven depth (PSO and DE dominate).
- Purely descriptive — no meta-analysis or quantitative synthesis.
- Theoretical claims (the 50% probability argument) accepted uncritically.
- No reproducibility assessment of surveyed papers.

5. Cross-Cutting Analysis

5.1 Where the Field Has Improved Over O-PSO

Three trends stand out across the studied papers:

1. **From initialization-only to full-search OBL.** The focal O-PSO uses OBL only in the first generation. Later papers (OVCPSO, OPSO-Cauchy, GOPSO, BPSO-OBL) apply OBL throughout the search via generation jumping with a jumping rate J_r. This consistently outperforms initialization-only OBL when J_r is properly chosen.
2. **Increasing statistical rigour.** Early papers report only means and standard deviations. Later work (BPSO-OBL, SDE-OBL, IOBL-PSO) adds Wilcoxon rank-sum or Friedman tests — a clear improvement in evidence quality.
3. **Broader baselines and benchmarks.** Early papers compare against PSO1, PSO2, and PPO on four classical functions. By 2015 the standard had moved to CEC 2005/2010/2013 suites and strong baselines such as CLPSO, FIPS, and GOPSO.

5.2 Open Problems That Persist

Several criticisms recur across nearly every paper:

- Jumping rate J_r is treated as a hyperparameter with no adaptive scheme.
- Computational cost is rarely reported in function evaluations; many papers compare iterations instead, hiding the cost of evaluating opposites.
- Real-world deployment is rare. Only Cao et al. (reactive power dispatch) and LatinPSO (ODE inference) use genuine real-world problems.
- Theoretical convergence analysis is almost entirely absent across the OBL-PSO line.

5.3 Summary Comparison Table

6. Which Paper Is Best, and Why

There is no single "best" answer that holds across every criterion, so we name a winner under each lens before giving an overall verdict.

6.1 Best by Foundational Importance

Tizhoosh (2005). Without this paper, none of the others exist. Every subsequent OBL-PSO, OBL-DE, OBL-AMO, GOBL, or quasi-opposite paper builds on the linear-opposite formulation introduced here. By citation count and field-defining impact, it is the most important paper of the set.

Paper	Year	Stat. tests	Real-world	Key contribution
Tizhoosh	2005	–	–	Introduced OBL itself.
O-PSO (focal)	2009	No	No	OBL at initialization.
OPSO-Cauchy	2007	No	No	OBL + Cauchy mutation.
OVCPSO	2009	No	No	OBL + velocity clamping.
OCLPSO	2008	No	No	OBL inside CLPSO exemplars.
GOPSO	–	Partial	No	Generalized OBL + Cauchy.
ODE	–	Some	No	OBL applied to DE.
SDE-OBL	–	Yes	No	OBL in shuffled DE; CEC 2005.
BPSO-OBL	2015	Yes	No	OBL on pbest + rebel learning.
OBPSO	2012	No	Partial	OBL + boundary search for constraints.
COPSO	–	Partial	Yes (FS)	OBL + conditional for feature selection.
IOBL-PSO	–	Yes	No	Refined OBL formulation.
WE-PSO	–	Partial	No	WELL-based initialization.
Cao et al. RPD	–	No	Yes	OBL for power-system control.
Kang et al.	–	No	No	OBL under noisy fitness.
Xu et al. Review	2014	n/a	n/a	Field-wide survey.

Table 1: High-level comparison of selected papers across the key evaluation dimensions.

6.2 Best by Methodological Rigour

Liu, Xu, Wang, Zeng (2015) — BPSO-OBL. This is the cleanest experimental work in the entire set. It applies non-parametric statistical tests (Wilcoxon), compares against seven competitors including strong modern baselines (CLPSO, GOPSO, FIPS), scales to 50 dimensions, and pairs a genuinely new mechanism (OBL on pbests plus a rebel-learning repulsive term derived from the worst pbest) with a thoughtful justification for omitting initialization-OBL.

6.3 Best by Practical Real-World Applicability

Cao, Ding, Wang, Chen — OBL for Reactive Power Dispatch. Almost every other paper lives on synthetic benchmarks. This one solves an actual engineering problem (reactive-power optimisation in a power system) with a realistic three-part cost model and tests on multiple real grids. Its experimental hygiene is weaker than BPSO-OBL, but it is the only paper that demonstrates OBL delivering value on a deployed problem.

6.4 Best by Reference Value

Xu et al. (2014) — OBL Review. If a new researcher entered the field today, this is the single document that gives the fastest, most coherent orientation across 138 papers and the OBL variant zoo.

6.5 Overall Verdict

Weighing all four criteria together, our overall winner is **BPSO-OBL (Liu, Xu, Wang, Zeng, 2015)**.

The focal O-PSO opened the door by showing that OBL can help PSO, but its experimental rigour is, by modern standards, weak (four benchmarks, no statistical tests, swarm size 20, no FE-budget analysis). The papers that came after fall into two camps: those that bolt extra mechanisms onto PSO without strong evidence, and those that genuinely move the needle. BPSO-OBL is the strongest in the second camp because:

- It targets the *root cause* of BPSO's premature convergence (poor pbest quality) rather than patching it with mutations or external jumps.
- Its rebel-learning term is theoretically motivated rather than purely heuristic.
- Its empirical evaluation is the most rigorous in the entire surveyed set — Wilcoxon tests, seven strong competitors, scaling to 50 dimensions.
- It is principled about *where* to apply OBL (explicitly omitting initialization-OBL with a budget argument), which is itself an instructive design lesson.

For *field-defining importance*, Tizhoosh (2005) remains unmatched. For *real-world deployment*, Cao et al. stands alone. For *quick orientation to the literature*, Xu et al. (2014) is the right starting point. But for *the strongest single piece of methodologically modern OBL-PSO research*, BPSO-OBL is our pick.

6.6 Where the Focal Paper Stands

The focal paper (O-PSO, 2009) is best understood as a stepping stone. It is conceptually clean and easy to extend, but methodologically thin by today's standards. Its lasting value is pedagogical: it is the simplest demonstration that OBL helps PSO, which is exactly the right starting point for a course project. The progression from O-PSO to OVCPSO to GOPSO to BPSO-OBL is, in microcosm, the maturation curve of an entire research subfield.

7. Conclusion

Across the four phases of this course project, the group examined a representative slice of the Opposition-Based Learning literature, anchored on the focal O-PSO paper of Jabeen, Jalil, and Baig (2009). The set spans the foundational OBL paper by Tizhoosh, multiple PSO variants that place OBL at initialization or throughout the search, extensions into Differential Evolution and Animal Migration Optimization, two domain-specific applications, and a field-wide survey.

Three observations stand out. First, OBL is most beneficial when applied throughout the search rather than only at initialization — a clear progression visible from O-PSO to BPSO-OBL. Second, methodological rigour has improved noticeably between 2005 and 2015, with later papers reporting non-parametric statistical tests and comparing against stronger baselines. Third, despite this progress, the field still has clear open problems: an adaptive jumping rate, theoretical convergence analysis, and fairer reporting of computational cost in function evaluations rather than iterations.

Within this set, our overall winner is **BPSO-OBL (Liu et al., 2015)** for its methodological rigour, principled design, and meaningful contribution beyond the prior art. Tizhoosh (2005) holds the title of most foundational paper, Cao et al. and Xu et al. are the strongest representatives of practical application and field-wide reference. The focal paper, O-PSO, retains its place as the cleanest pedagogical entry point to the entire research line — which is precisely what made it the right anchor for this project.